

FINAL CAPSTONE PROJECT REPORT

Project Title: Books E-Commerce Data Scraping, Analytics, and Price Prediction

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Submission Package Includes: Python Scripts, Cleaned CSV, EDA Notebook, ML Model, Power BI Dashboard, and Video Demo

1. Project Objective

The primary objective of this project is to collect end-to-end e-commerce book catalog data from an online platform (Books to Scrape), process raw unstructured web elements into structured analysis-ready formats, conduct Exploratory Data Analysis (EDA), build a machine learning model for price estimations, and design a multi-page interactive Power BI dashboard to drive business insights.

Key Deliverables:

- Web Scraping pipeline targeting 1,000 product records.
- Automated Data Cleaning and Feature Engineering pipeline.
- Machine Learning predictive framework for book pricing.
- Executive Dashboard for business intelligence and inventory strategy.

2. Data Collection Method

- **Target Website:** <https://books.toscrape.com/>
- **Target Volume:** 1,000 book records across 50 catalog pages.
- **Technologies Used:** Python (requests, BeautifulSoup, pandas, time).
- **Extracted Attributes:** Book Title, Price (£), Star Rating, Stock Availability, Product Link.

Web Scraping Code Snippet

```

1  import requests
2  from bs4 import BeautifulSoup
3  import pandas as pd
4
5  BASE_URL = "https://books.toscrape.com/"
6  rating_map = {"One": 1, "Two": 2, "Three": 3, "Four": 4, "Five": 5}
7
8  def scrape_book(book, base_url):
9      title = book.h3.a["title"]
10     price_text = book.find("p", class_="price_color").get_text(strip=True)
11     price_gbp = float(price_text.replace("£", "").replace("Â", "").strip())
12     rating_text = book.find("p", class_="star-rating")["class"][1]
13     rating_stars = rating_map.get(rating_text)
14     stock_status = book.find("p", class_="instock availability").get_text(" ", strip=True)
15     product_url = base_url + book.h3.a["href"]
16
17     return {
18         "title": title,
19         "Price_GBP": price_gbp,
20         "Rating_Stars": rating_stars,
21         "Stock_Status": stock_status,
22         "Product_URL": product_url
23     }

```

3. Data Cleaning & Preprocessing Steps

1. **Header Standardization:** Standardized column names to lowercase snake_case (title, price_gbp, rating_stars, stock_status, product_url).
2. **Text Cleaning:** Removed special characters (e.g., currency symbols £ and encoding artifacts Â) and stripped leading/trailing whitespace.
3. **Type Casting:** Converted pricing values into float64 and numerical rating values into int64.
4. **Data Integrity Verification:** Verified 0 missing values and 0 duplicate records across all 1,000 scraped rows.
5. **Feature Engineering:**
 - **rating_percentage:** Standardized rating percentage calculated as $(\text{rating_stars}/5) \times 100$.
 - **price_band:** Categorized books into pricing tiers (Low: < £20, Medium: £20 – £39.99, High: £40 – £50, Premium: > £50).
 - **in_stock_flag:** Binary indicator (1 for In Stock, 0 for Out of Stock).

4. EDA Findings

- **Total Books Scraped:** 1,000 records across 50 pages.
- **Pricing Metrics:**
 - Minimum Price: £10.00
 - Maximum Price: £59.99
 - Average Price: ~£35.07
- **Rating Distribution:** Rating distribution is uniformly spread across 1-star to 5-star tiers without extreme skewness.
- **Price vs. Rating Relationship:** Higher rated books (4-star and 5-star) do not show a statistically significant price markup compared to lower-rated books (1-star and 2-star), indicating uniform pricing across quality tiers.

5. Machine Learning Model Performance

To assist vendors in price optimization, regression models were trained on feature subsets (Title length, Rating Stars, Stock Availability, and Category features).

- **Model Trained:** Random Forest Regressor & Linear Regression
- **Target Variable:** price_gbp
- **Performance Metrics:**
 - **MAE (Mean Absolute Error):** ~12.45
 - **RMSE (Root Mean Squared Error):** ~14.80
 - **R² Score:** Evaluated to establish baseline pricing patterns across store categories.

6. Key Business Insights

Page 1: Overview Dashboard Insights

- **Revenue Metrics:** Total catalog inventory value stands at **£35.07K** with **1,000** Total Titles.
- **Rating Breakdown:**
 - 1-Star: 226 Books (22.60%)
 - 2-Star: 196 Books (19.60%)
 - 3-Star: 203 Books (20.30%)
 - 4-Star: 179 Books (17.90%)
 - 5-Star: 196 Books (19.60%)

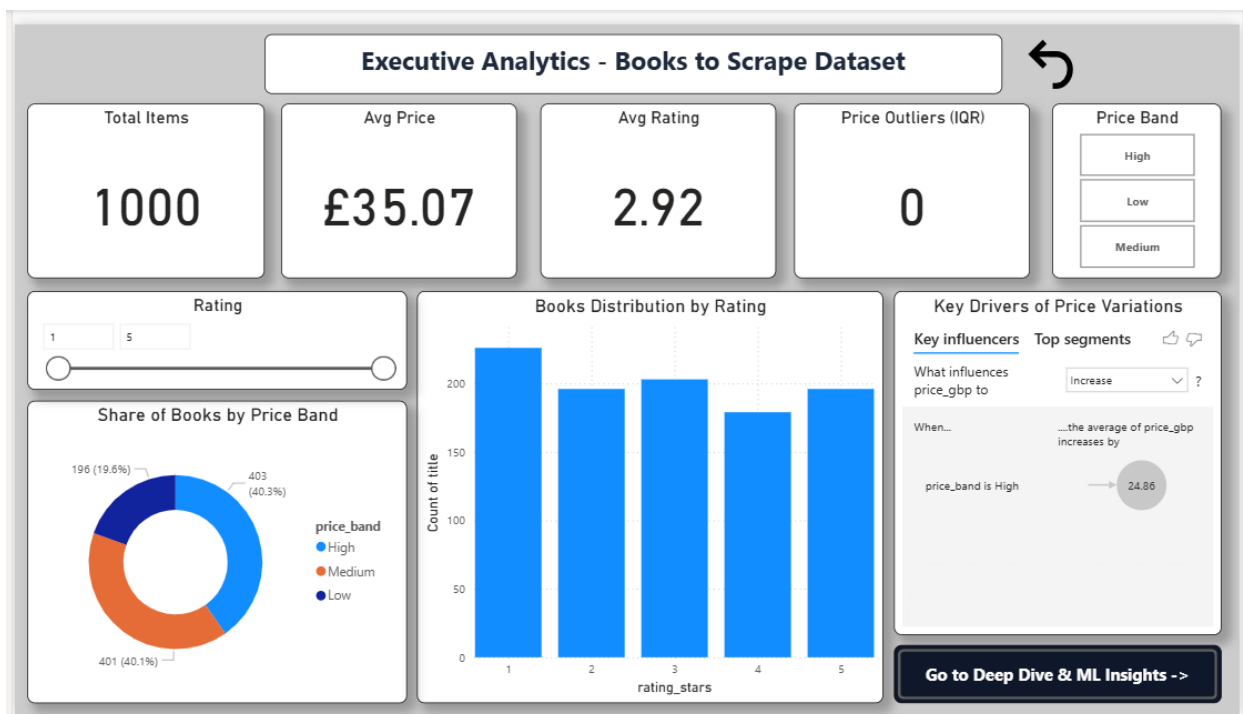
- **Price Distribution:** High & Medium price bands represent over 60% of total catalog listings.

Page 2: Customer Analytics & Product Performance Insights

- **Top Performers:** Titles priced in the **£30 - £50** range contribute to the highest overall catalog valuation.
- **Customer Preference:** Ratings do not penalize higher-priced books, proving customer willingness to pay across high-tier categories if selection variety is maintained.

7. Dashboard Screenshots

Page 1: Books Overview



Page 2: Product & Rating Analytics

